



EAST WEST UNIVERSITY

LAB REPORT 7

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Course Name: Electronic Circuits

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Group No: 10A

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Title:**To study the Performance of a Summing Amplifier****Objective:**

1. To study the operation of an op-amp based summing amplifier.
2. To observe the summation of DC and AC input signals.
3. To compare input and output waveforms using an oscilloscope.
4. To verify the practical performance with different input waveforms.

Theory:

A summing amplifier is a circuit made using an operational amplifier (Op-Amp) that adds two or more input voltages and produces a single output voltage. In this experiment, a 741 Op-Amp was used to study how different input signals are combined at the output.

The circuit used is an inverting summing amplifier, where all input signals are connected to the inverting (-) terminal of the Op-Amp through separate resistors. The non-inverting (+) terminal is connected to ground. A feedback resistor is connected between the output and the inverting terminal to provide negative feedback and keep the amplifier stable.

The output voltage is equal to the negative sum of all the input voltages. This means the output is 180° out of phase (inverted) with respect to the combined input signals.

The output voltage of an inverting summing amplifier is:

$$V_{\text{out}} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \dots + \frac{V_n}{R_n} \right)$$

If all the resistors have the same value ($R_1=R_2=R_f$)($R_1=R_2=R_f$), the equation becomes:

$$V_{\text{out}} = -(V_1 + V_2)$$

This means the output is simply the negative of the sum of the input voltages.

During the experiment, different types of input waveforms such as square wave, triangular wave, and sine wave were applied. The oscilloscope was used to compare the input and output signals. It was observed that the output waveform followed the combined input signals but appeared inverted. Different input voltages also

produced different output amplitudes, showing that the summing amplifier correctly adds the input signals.

Summing amplifiers are widely used in audio mixers, signal processing circuits, analog computers, measurement systems, and data acquisition systems, where multiple signals need to be combined into a single output.

Components and Apparatus:

1. 741 Op-Amp
2. Resistors: $2.2\text{ k}\Omega$ (3 pcs), $10\text{ k}\Omega$ (1 pc)
3. Power Supply ($\pm 12\text{ V}$)
4. Function Generator
5. Oscilloscope
6. Breadboard
7. Connecting Wires

Working Diagrams:

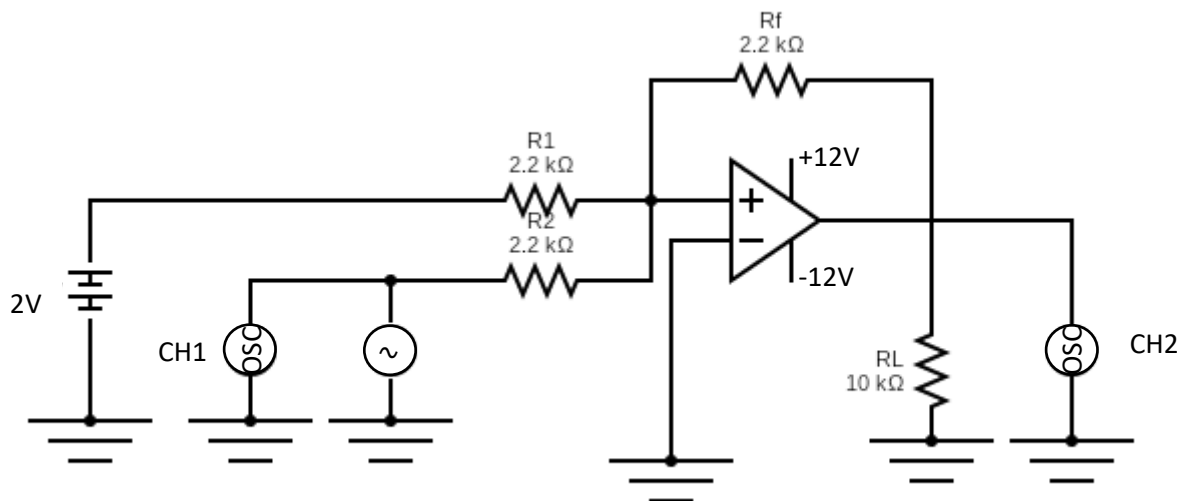


Figure 1: Summing Amplifier

Observation

Different waveforms (square, triangular and sinusoidal) were applied to verify the operation of the summing amplifier. The oscilloscope showed that the output remained inverted with respect to the summed input while the frequency stayed close to 1 kHz. Different DC offsets and input amplitudes produced corresponding changes in the output voltage, confirming the expected summing action.

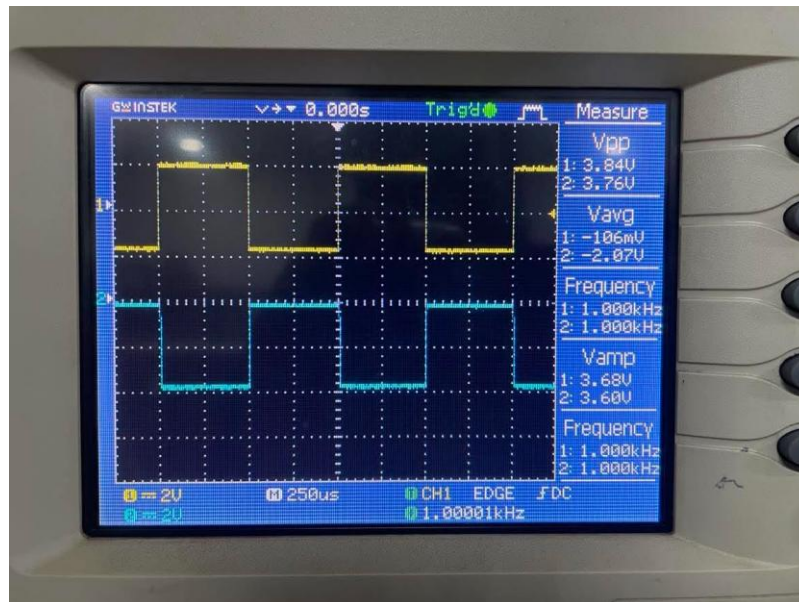


Figure 2: Square-wave summing response

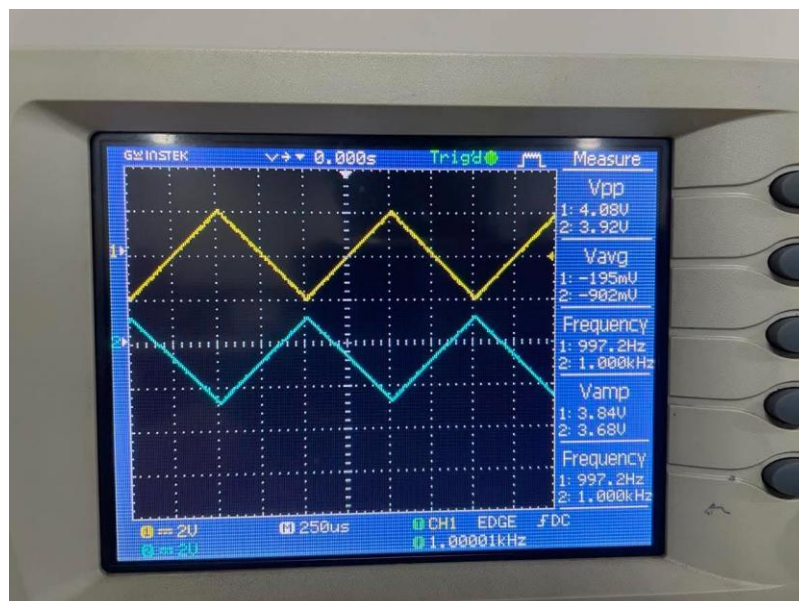


Figure 3: Triangular-wave summing response.

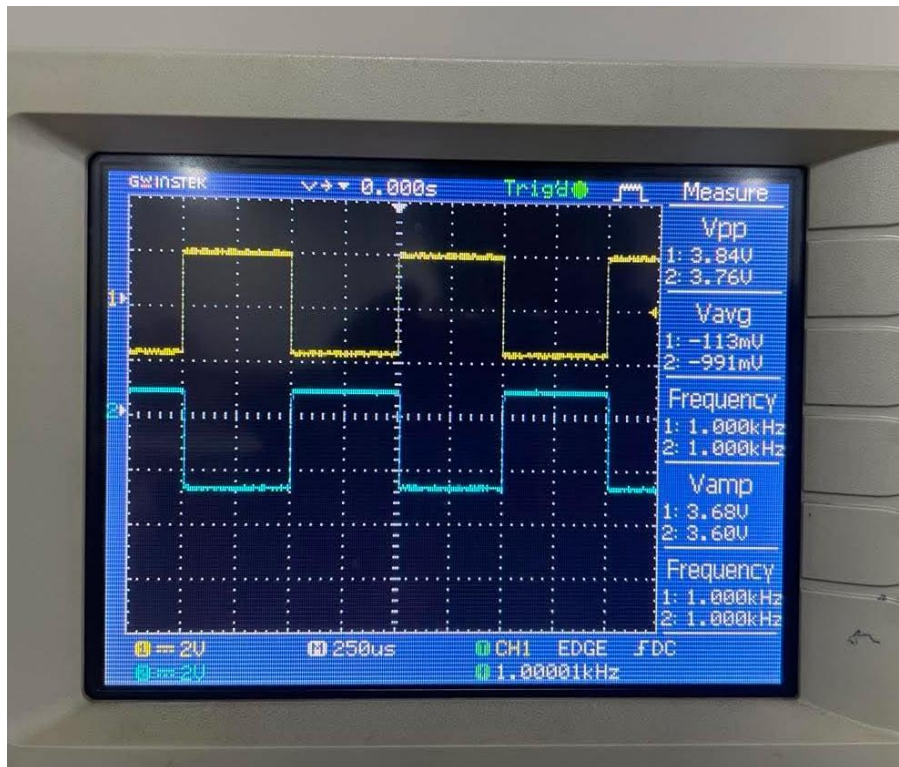


Figure 4: Square-wave summing response.

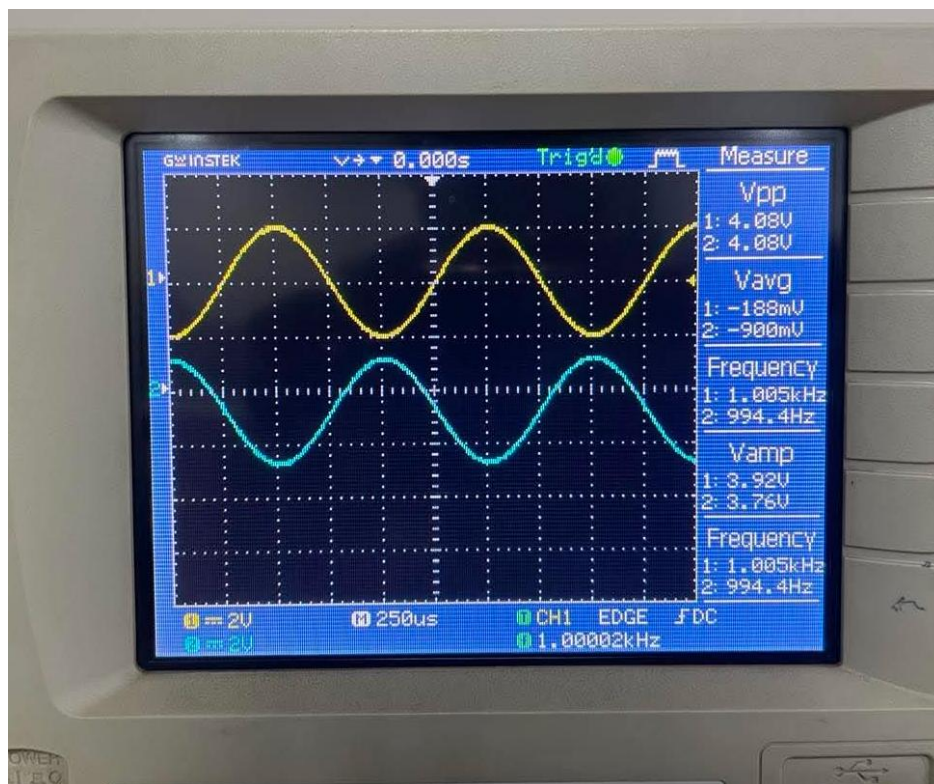


Figure 5: Sinusoidal response.

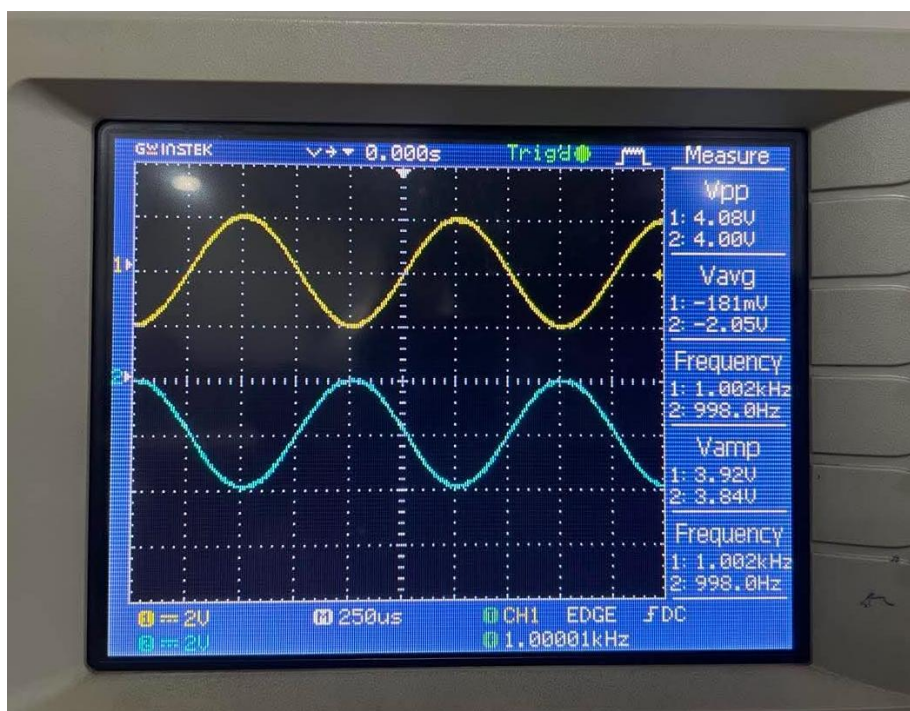


Figure 6: Sinusoidal response with DC offset.

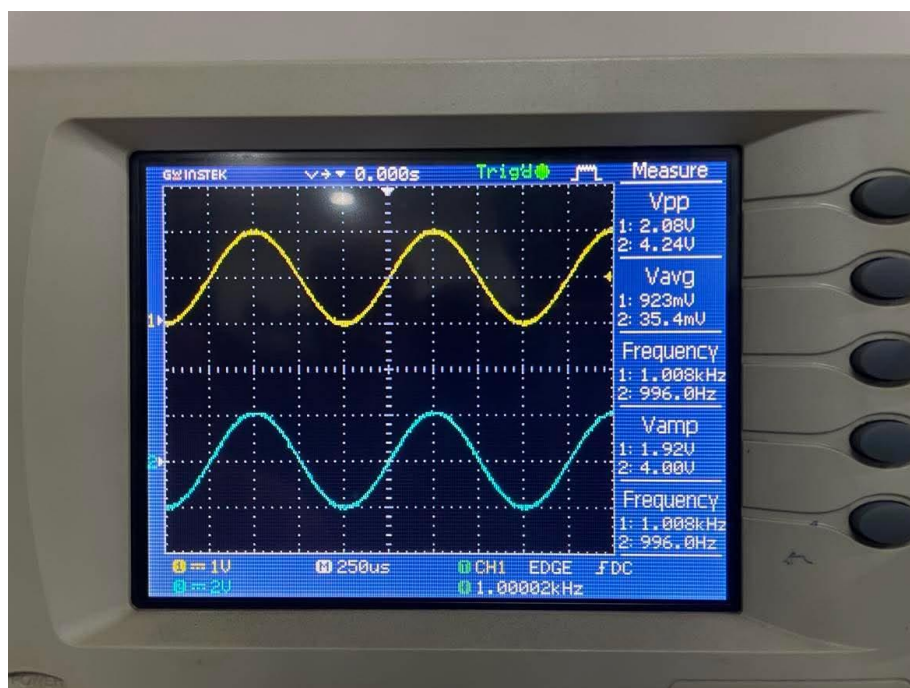


Figure 7: Summing amplifier output for different input amplitude

Result

The summing amplifier operated successfully. The observed output waveforms matched the expected inverted sum of the applied input signals for different waveform types and voltage levels.

Discussion:

In this experiment, the performance of a 741 operational amplifier configured as an inverting summing amplifier was studied. The circuit was built on a breadboard using three $2.2\text{ k}\Omega$ resistors, one $10\text{ k}\Omega$ load resistor, a $\pm 12\text{ V}$ power supply, a function generator, and an oscilloscope.

One input of the circuit was connected to a 2 V DC source, while the other input received different AC signals from the function generator. Several waveforms, including square wave, triangular wave, and sine wave, were applied at approximately 1 kHz . The input and output signals were observed on the oscilloscope.

Six different oscilloscope readings were recorded by changing the waveform type and input voltage. These readings show that the amplifier successfully added the input signals and produced a single output waveform. The output waveform remained inverted because the circuit was configured as an inverting summing amplifier. The frequency remained close to 1 kHz , while the output voltage changed according to the applied input signals.

The experiment confirmed that the summing amplifier worked correctly for different waveform shapes and voltage levels. The practical results matched the expected behavior of an inverting summing amplifier, proving that the circuit can combine multiple signals into one output efficiently.